Synthesis and Characterisation of Ferrocene

CM3191 • 10 and 17 September 2026 • Instructor: Dr Edith Chan
KIM DONGHWAN (A0266094X) • Thursday session, fume hood B7 (SN9) • e1075536@u.nus.edu

1. Aims

- To prepare ferrocene, $Fe(\eta^5\text{-}C_5H_5)_2$, from cyclopentadiene, KOH and $FeCl_2 \cdot 4\,H_2O$ under nitrogen.[1]
- To purify the crude product by sublimation.
- To characterise the product by stereomicroscopy, ^{1}H NMR, ATR-IR, UV–Vis, ESI-MS and magnetic susceptibility.
- To compare the results with the sandwich structure and the 18-electron, low-spin d^6 configuration of ferrocene.

2. Results and Discussion

2.1 Synthesis, yield and purification

Cyclopentadiene (1.0 mL, 0.80 g, 12.10 mmol) was deprotonated with KOH (4.0170 g, 71.6 mmol) in anhydrous DME (12 mL) under nitrogen. $FeCl_2 \cdot 4\,H_2O$ (1.3274 g, 6.677 mmol) in DMSO (15 mL) was added dropwise. The mixture was quenched on ice with concentrated HCl, and the crude product was filtered, washed with water and dried under vacuum.[1]

$$2\,C_5H_6 + 8\,KOH + FeCl_2 \cdot 4\,H_2O \longrightarrow Fe(C_5H_5)_2 + 2\,KCl + 6\,KOH \cdot H_2O \tag{1}$$

Excess KOH deprotonates cyclopentadiene (pK_a 18), and two $C_5H_5^-$ ions then bind Fe^{2+}. Cyclopentadiene is limiting, with iron in 10% excess, and the crude product (0.0699 g) corresponds to a 6.2% yield (Table 1). After 10 min the cyclopentadienide solution was pinkish brown rather than deep burgundy, which suggests incomplete deprotonation. The green colour on quenching is expected from the excess Fe^{2+}.

Sublimation at 100–140 °C onto an ice-cooled cover gave 0.0526 g of orange crystals, a 75% recovery and a 4.7% isolated yield. After samples had been taken for microscopy, NMR, ESI-MS and IR, only 9 mg remained, too little for UV–Vis and the magnetic balance. A second portion of crude (0.3021 g) was therefore sublimed (0.1805 g, 60% recovery) for these two measurements, so the yield refers to the original synthesis only. Because the crude yield was only 6.2%, most of the loss occurred before sublimation.

Table 1. Masses, yields and sample use (analytical balance, ±0.0001 g; Appendix E).

Quantity	Mass / g	Yield or recovery	Used for
Theoretical ferrocene (CpH limiting)	1.1258	—	
Crude ferrocene (1.6127 − 1.5428)	0.0699	6.2%	sublimation, microscopy
Sublimate from the original crude (cycles 1, 2)	0.0526	75.3% rec.; 4.7% overall	microscopy, NMR, ESI-MS, IR
Second crude portion	0.3021	—	
Sublimate from the second portion (cycles 3, 4, dish)	0.1805	59.7% rec.	UV–Vis (0.0232 g), MSB (0.0867 g)

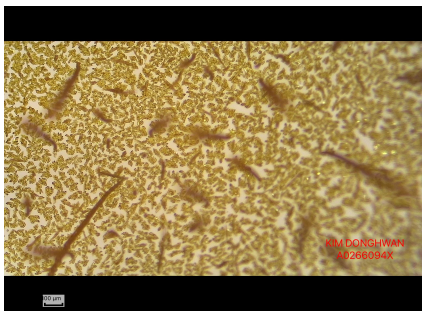
Under the stereomicroscope, the crude is a compact mass of yellow–orange granules without defined faces (Figure 1a), whereas both sublimates consist of discrete orange plates and needles with sharp edges (Figure 1b,c). This change in habit is consistent with recrystallisation by sublimation. Because the two sublimates look alike, the UV–Vis and magnetic results were interpreted assuming that both have the same composition, although appearance alone cannot confirm this.

2.2 ^{1}H NMR spectroscopy

The ^{1}H NMR spectrum of the sublimate (about 1 mg in 0.6 mL $CDCl_3$) shows one sharp singlet at δ 4.17 for the ten ring protons (Figure 2; lit. 4.15).[2] Because both η^5-rings are equivalent and rotate freely, all ten C–H protons are identical on the NMR time scale. The weak lines at δ 4.39 and 3.95 are ^{13}C satellites ($^1J_{CH}$ = 175 Hz) rather than impurities, since together they integrate to about 1% of the singlet, as expected from the natural abundance of ^{13}C (Figure 5). No signals from dienes (δ 5–7), DME (δ 3.40, 3.55) or DMSO (δ 2.62) were detected.[3] Water (δ 1.55) and acetone (δ 2.17), probably from the solvent and the acetone-rinsed glassware, appear together with a small unassigned aliphatic signal at δ 1.25.



(a) crude, Session I



(b) sublimed, original batch (100 μm bar)



(c) sublimed, second portion (100 μm bar)

Figure 1. Stereomicrographs at the manual's magnification. (a) Crude ferrocene. (b) Sublimate of the original crude, used for NMR, ESI-MS and IR. (c) Sublimate of the second portion, used for UV–Vis and the magnetic balance. A higher-magnification image of (c) is shown in Figure 4.

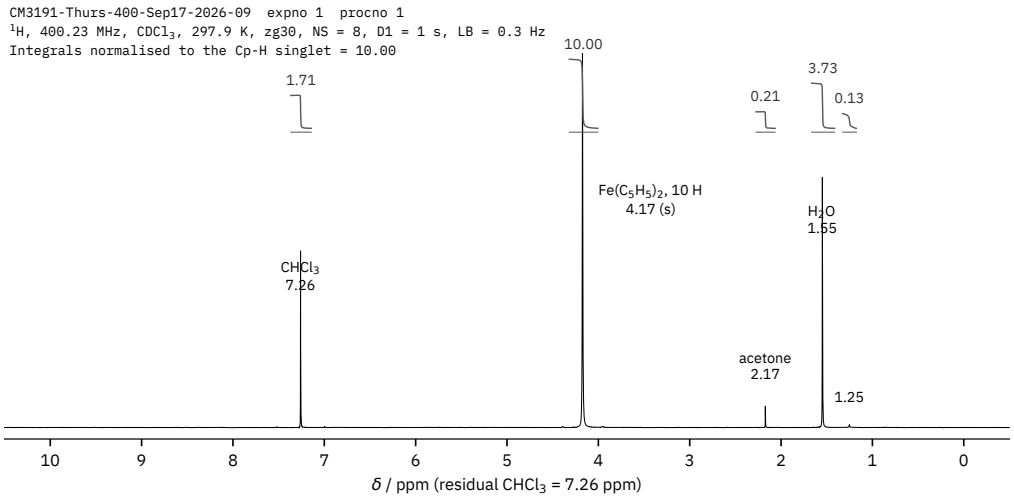


Figure 2. ^{1}H NMR spectrum of the sublimed ferrocene (400.23 MHz, $CDCl_3$, 297.9 K, zg30, 8 scans, D1 = 1 s), referenced to $CHCl_3$ at 7.26 ppm. Integrals are relative to Cp-H = 10.00. Expansions are in Figure 5.

2.3 ATR-IR spectroscopy

The ATR-IR spectrum (Figure 6) shows five characteristic ferrocene bands, each within 10 cm^{-1} of the values assigned by Lippincott and Nelson.[4] They are the ring C–H stretch at 3095 cm^{-1} (lit. 3085), the antisymmetric C–C stretch at 1407 (1411), ring breathing at 1104 (1108), the in-plane C–H bend at 999 (1002) and the out-of-plane C–H bend at 812 cm^{-1} (811), which is the strongest band. Neither an aliphatic C–H stretch below 3000 cm^{-1} nor a broad O–H band is present. Weak bands at 1631–1775, 1255, 1190, 1047, 855 and 786 cm^{-1} may be overtone or combination bands, but a minor impurity cannot be ruled out.

2.4 UV–Vis spectroscopy and ligand-field assignment

A solution of the second sublimate (0.0232 g in 25.00 mL methanol, 4.988×10^{-3} M) was measured undiluted in a 1.00 cm cell (Figure 7). The maxima were at 440.5 nm ($A = 0.416$) and 325.0 nm ($A = 0.239$), both within the 0.1–1.0 absorbance range.

$$\varepsilon_{440} = \frac{0.416}{4.988 \times 10^{-3} \times 1.00} = 83\ \mathrm{M^{-1}\,cm^{-1}}, \qquad \varepsilon_{325} = \frac{0.239}{4.988 \times 10^{-3} \times 1.00} = 48\ \mathrm{M^{-1}\,cm^{-1}} \tag{2}$$

A handwritten version of this calculation is on the signed UV–Vis printout (Appendix D). Sohn, Hendrickson and Gray report bands at 440.5 nm (ε 91) and 324.7 nm (ε 49) in EPA.[5,6] While the band positions agree within 1 nm, the absorptivities are 8% and 2% lower. Values below 100 M^{-1} cm^{-1} indicate Laporte-forbidden d–d transitions. In the assignment of Sohn *et al.*, with the $^1A_{1g}$, $(e_{2g})^4(a_{1g})^2$ ground state in D_{5d}, the 440 nm envelope contains the $^1A_{1g} \rightarrow a\,^1E_{1g}$ and $^1A_{1g} \rightarrow {}^1E_{2g}$ transitions, and the 325 nm band is $^1A_{1g} \rightarrow b\,^1E_{1g}$.[5] All three move an electron from the filled, mainly metal e_{2g} and a_{1g} orbitals into the

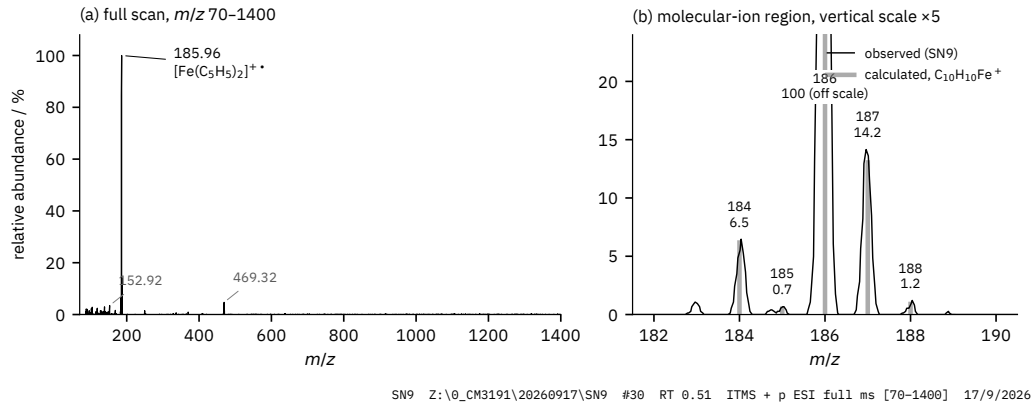


Figure 3. Positive-ion ESI mass spectrum of the sublimed ferrocene (sample SN9; Thermo LTQ XL ion trap, flow injection in methanol, capillary 43 V).[8] (a) Full scan, m/z 70–1400. (b) Molecular-ion region with the envelope calculated for $C_{10}H_{10}Fe^+$ (grey bars). The original printout is in Appendix D.

empty antibonding e^*_{1g} level. A non-absorbing impurity would lower both absorptivities equally, so the different shortfalls more likely reflect the solvent (methanol here, EPA in the literature) or the baseline.

2.5 ESI-MS

The positive-ion ESI spectrum of the sublimate (Figure 3) shows one dominant ion at m/z 185.96, and no other ion exceeds 5%. Because ferrocene is easily oxidised and the electrospray source can act as an electrolytic cell,[7] this ion is assigned to the ferrocenium radical cation $[Fe(C_5H_5)_2]^{+\bullet}$. No $[M + H]^+$ or $[M+Na]^+$ ion (m/z 209) appears, although sodium is present in the system.[8] The peaks at m/z 184, 185, 187 and 188 (6.5, 0.7, 14.2 and 1.2%) match the envelope calculated for $C_{10}H_{10}Fe^+$ within about one percentage point (Appendix C).[9] No dimer (m/z 373) or Cp-loss fragment (m/z 121) was detected, and the spectrum supports the formula $C_{10}H_{10}Fe$.

2.6 Magnetic susceptibility

A ground sample of the second sublimate (0.0867 g, packed to 17 mm) was measured on a Sherwood MSB AUTO at 23 °C (Figure 8).[8] The display read $\chi_v = -0.005 \times 10^{-4}$ and $\chi_g = 0.000 \times 10^{-4}$ cm^3 g^{-1}. One unpaired electron would give a display of about $+0.068 \times 10^{-4}$ at 296 K, but neither value is positive, so the data give no sign of paramagnetism. However, the literature value for ferrocene, $\chi_M = -125 \times 10^{-6}$ cm^3 mol^{-1},[6] corresponds to a χ_g display of -0.007×10^{-4}, and the volume reading implies -0.003×10^{-4} or lower, so the two readings disagree with each other and with the literature (Appendix B). No repeat zero, blank or calibration standard was measured, so no susceptibility or moment is derived. The packed mass was below the manual's 0.10–0.12 g and cannot be checked, because the two tare values on the datasheet disagree.

2.7 Sources of error

Possible losses in the reaction include incomplete deprotonation, re-dimerisation and evaporation of cyclopentadiene, air oxidation of Fe^{2+} during the slow addition and addition of less than the intended 1.0 mL of cyclopentadiene. Further product was left on glassware and filters or lost when the sublimation cover was lifted, although none of these contributions was measured. Water or KCl in the crude would raise the crude yield and lower the apparent recovery.

3. Conclusion

Ferrocene was synthesised from cyclopentadiene and $FeCl_2 \cdot 4\,H_2O$ via potassium cyclopentadienide and purified by sublimation in 4.7% isolated yield (6.2% crude, sublimation recoveries 75% and 60%), with most of the loss occurring before sublimation. For the original sublimate, the ^{1}H NMR singlet at δ 4.17 with its ^{13}C satellites, five characteristic IR bands and the ferrocenium ion at m/z 186 with its isotope pattern identify the product as ferrocene. Minor solvent signals and weak unassigned NMR and IR signals were also observed. The second sublimate showed ligand-field bands at 440.5 nm ($\varepsilon = 83$) and 325.0 nm ($\varepsilon = 48$), within 1 nm of the literature. The magnetic results were inconclusive. The spectroscopic data are consistent with the closed-shell, low-spin d^6 configuration of ferrocene.

References

[1] Chan, E. *CM3191 Chemical Synthesis Experiments: Lab Manuals for Inorganic Experiments*, version 0626; National University of Singapore, AY2026/27; Experiment 2, pp 12–22; Appendix (MSB AUTO), p 23.

[2] Powers, T. M. Structure of Ferrocene. *JoVE Science Education: Inorganic Chemistry*; JoVE: Cambridge, MA, 2017. `https://app.jove.com/v/10347` (accessed 8 October 2026).

[3] Fulmer, G. R.; Miller, A. J. M.; Sherden, N. H.; Gottlieb, H. E.; Nudelman, A.; Stoltz, B. M.; Bercaw, J. E.; Goldberg, K. I. NMR Chemical Shifts of Trace Impurities: Common Laboratory Solvents, Organics, and Gases in Deuterated Solvents Relevant to the Organometallic Chemist. *Organometallics* **2010**, *29*, 2176–2179.

[4] Lippincott, E. R.; Nelson, R. D. The Vibrational Spectra and Structure of Ferrocene and Ruthenocene. *Spectrochim. Acta* **1958**, *10*, 307–329.

[5] Sohn, Y. S.; Hendrickson, D. N.; Gray, H. B. Electronic Structure of Metallocenes. *J. Am. Chem. Soc.* **1971**, *93*, 3603–3612.

[6] Sohn, Y. S. *Electronic Structures of Metallocene Complexes*. Ph.D. Dissertation, California Institute of Technology, 1971; Part I, Table II; Part II, Table I (χ_M after Engelmann, *Z. Naturforsch.* **1953**, *8b*, 775). DOI: 10.7907/pg0q-y417.

[7] Van Berkel, G. J.; Zhou, F. Characterization of an Electrospray Ion Source as a Controlled-Current Electrolytic Cell. *Anal. Chem.* **1995**, *67*, 2916–2923.

[8] Chan, E. CM3191 Inorganic Canvas announcements, 24 September 2026 (ESI-MS instrumental settings) and 26 September 2026 (temperature in the magnetic measurement). National University of Singapore.

[9] Coursey, J. S.; Schwab, D. J.; Tsai, J. J.; Dragoset, R. A. *Atomic Weights and Isotopic Compositions*, version 4.1; National Institute of Standards and Technology: Gaithersburg, MD, 2015. `https://physics.nist.gov/Comp` (accessed 7 October 2026).

Appendix A. Supplementary figures



Figure 4. Higher-magnification stereomicrograph of the sublimate of the second portion (60 μm bar).

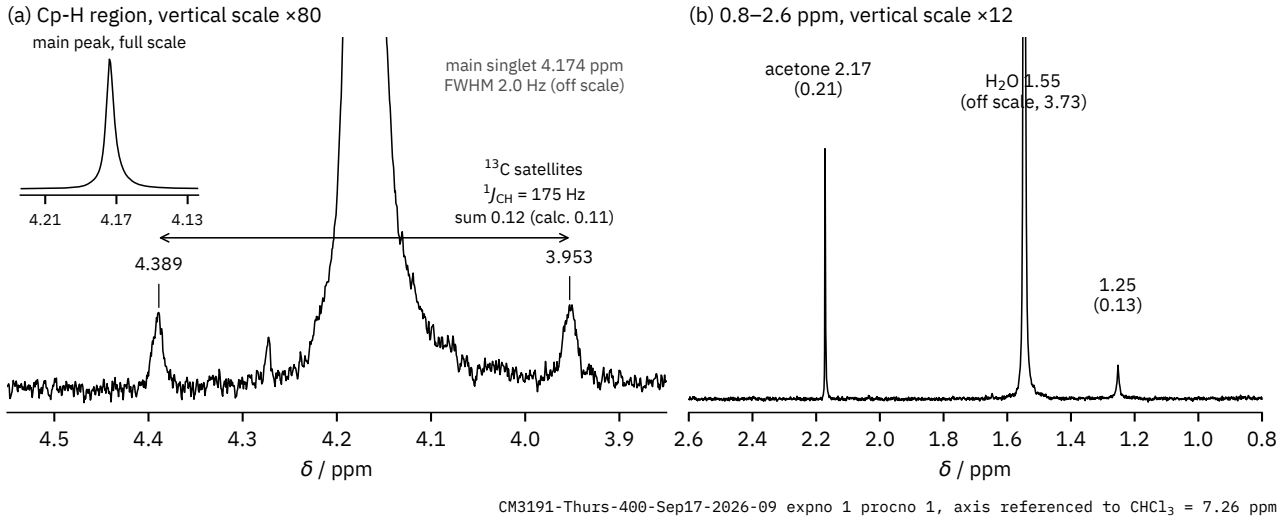


Figure 5. Expansions of Figure 2. (a) Cp-H region at 80-fold vertical gain showing the ^{13}C satellites ($^{1}J_{\mathrm{CH}}$ = 175 Hz), with the parent singlet at full scale inset. (b) 0.8–2.6 ppm at 12-fold gain, with integrals in brackets (Cp-H = 10.00).

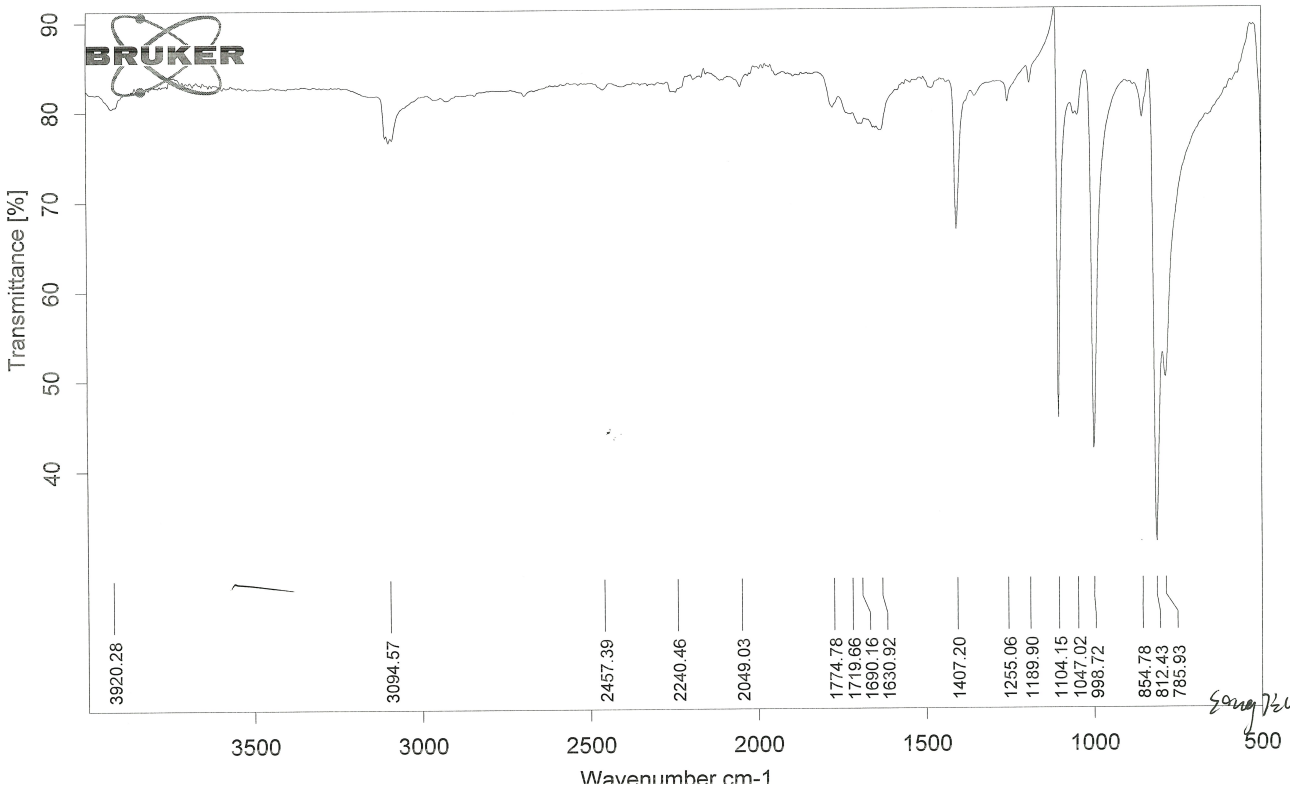


Figure 6. ATR-IR spectrum of the sublimed ferrocene (500–4000 cm^{-1}). The complete signed printout is in Appendix D.

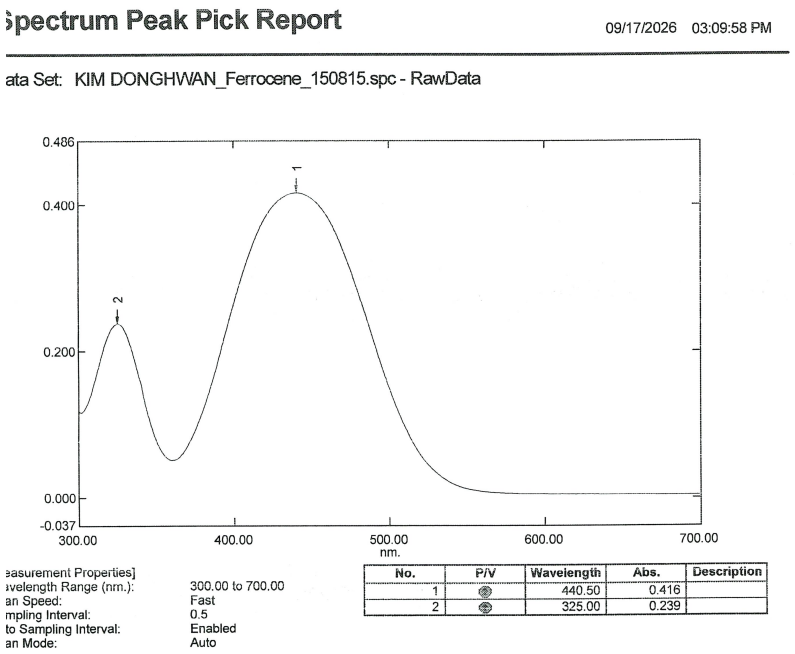
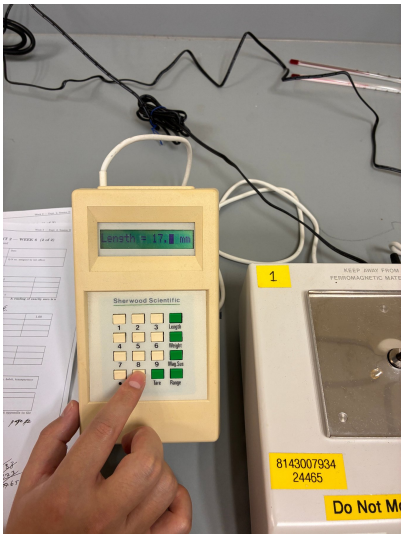


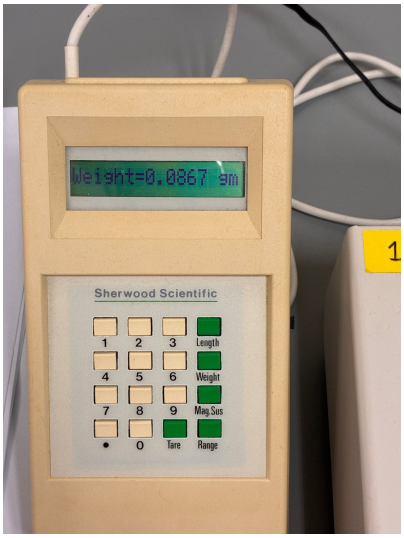
Figure 7. UV–Vis spectrum of the sublimed ferrocene (4.988×10^{-3} M in methanol, 1.00 cm) with the instrument peak table. The signed printout with the handwritten ε calculation is in Appendix D.



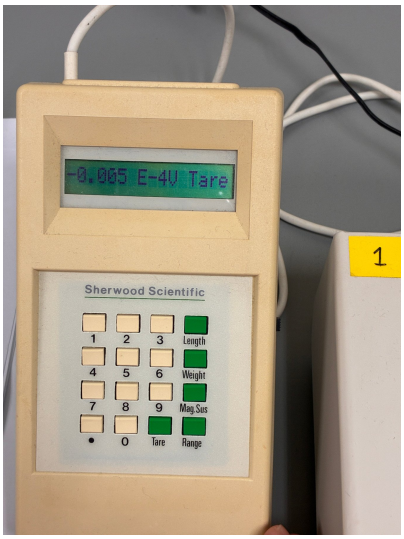
(1) zero reading before the tube was placed



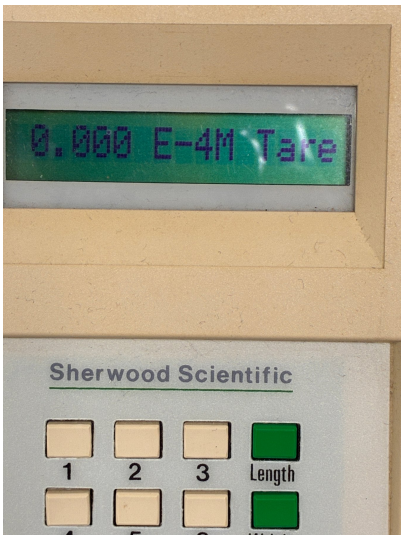
(2) length entry before the last digit, `17._ mm`



(3) weight entry, `Weight=0.0867 gm`



(4) volume susceptibility, `-0.005 E-4V`



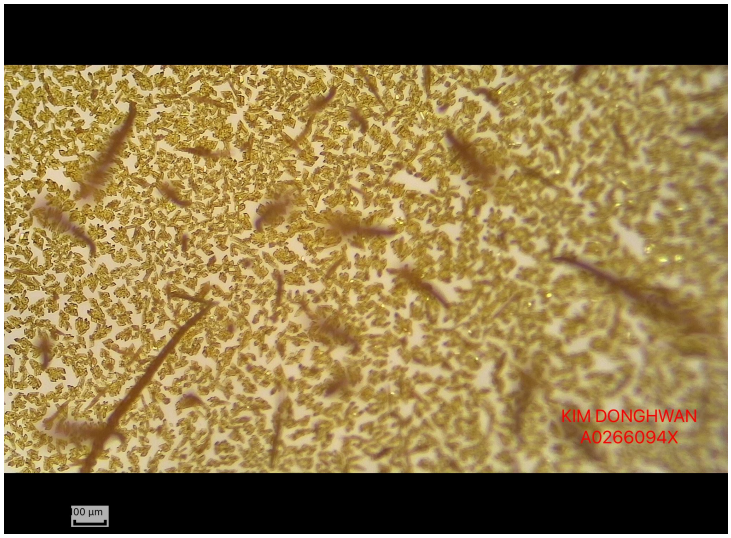
(5) mass susceptibility, `0.000 E-4M`

Figure 8. Sherwood MSB AUTO measurement (unit 1), in the order of the procedure. The zero reading in (1) was zero, but the photograph caught the display while it was refreshing. Weighing and packed-length photographs are in Appendix F.

Figures 1 and 2 are reproduced below at full size.



(a) crude, Session I



(b) sublimate of the original crude (100 μm bar)



(c) sublimate of the second portion (100 μm bar)

Figure 9. Stereomicrographs at the manual's magnification (full size of Figure 1).

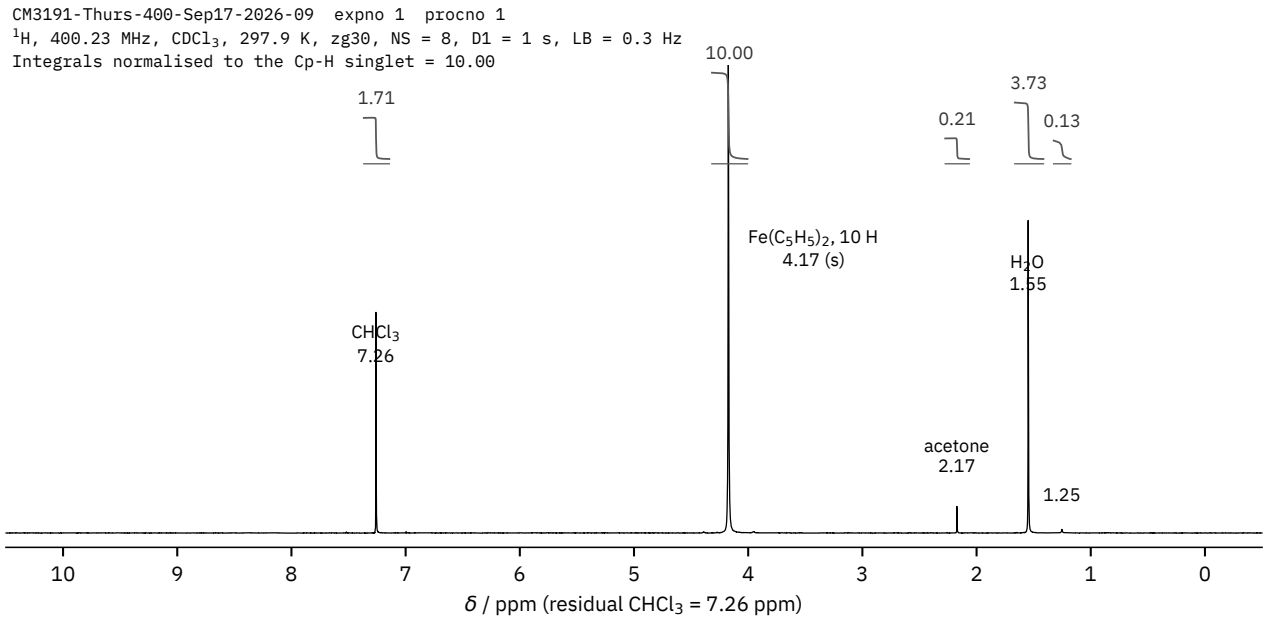


Figure 10. ^{1}H NMR spectrum of the sublimed ferrocene (full size of Figure 2; 400.23 MHz, $CDCl_3$, referenced to $CHCl_3$ at 7.26 ppm).

Appendix B. Calculations

Limiting reagent and theoretical yield.

$$n(\mathrm{C_5H_6}) = \frac{1.0\ \mathrm{mL} \times 0.80\ \mathrm{g\,mL^{-1}}}{66.10\ \mathrm{g\,mol^{-1}}} = 12.10\ \mathrm{mmol}, \qquad n(\mathrm{FeCl_2 \cdot 4\,H_2O}) = \frac{1.3274}{198.81} = 6.677\ \mathrm{mmol}, \tag{3}$$

$$n(\mathrm{KOH}) = \frac{4.0170}{56.11} = 71.6\ \mathrm{mmol\ (8\ equiv = 48.4\ mmol)}, \qquad n(\mathrm{Fc})_{\mathrm{theor}} = \tfrac{1}{2}\, n(\mathrm{C_5H_6}) = 6.051\ \mathrm{mmol}, \tag{4}$$

$$m_{\mathrm{theor}} = 6.051 \times 10^{-3} \times 186.04 = 1.1258\ \mathrm{g} \quad \text{(CpH limiting, Fe in 10\% excess)}. \tag{5}$$

Yields and recoveries.

$$m_{\mathrm{crude}} = 1.6127 - 1.5428 = 0.0699\ \mathrm{g}, \qquad Y_{\mathrm{crude}} = 0.0699/1.1258 = 6.21\%, \tag{6}$$

$$m_{\mathrm{subl,orig}} = 0.0403 + 0.0123 = 0.0526\ \mathrm{g}, \qquad Y_{\mathrm{isolated}} = 0.0526/1.1258 = 4.67\%, \tag{7}$$

$$R_{\mathrm{subl,orig}} = 0.0526/0.0699 = 75.3\%, \qquad m_{\mathrm{residue}} = 1.1083 - 1.0989 = 0.0094\ \mathrm{g}, \tag{8}$$

$$m_{\mathrm{subl,2nd}} = 0.1104 + 0.0563 + 0.0138 = 0.1805\ \mathrm{g}, \qquad R_{\mathrm{subl,2nd}} = 0.1805/0.3021 = 59.7\%. \tag{9}$$

Molar absorptivity (undiluted stock, $D = 1$).

$$c = \frac{0.0232\ \mathrm{g}}{186.04\ \mathrm{g\,mol^{-1}} \times 0.02500\ \mathrm{L}} = 4.988 \times 10^{-3}\ \mathrm{M}, \tag{10}$$

$$\varepsilon_{440} = \frac{0.416}{4.988 \times 10^{-3} \times 1.00} = 83.4\ \mathrm{M^{-1}\,cm^{-1}}, \qquad \varepsilon_{325} = \frac{0.239}{4.988 \times 10^{-3} \times 1.00} = 47.9\ \mathrm{M^{-1}\,cm^{-1}}, \tag{11}$$

$$\varepsilon_{440}/\varepsilon_{440,\mathrm{lit}} = 83.4/91 = 0.92, \qquad \varepsilon_{325}/\varepsilon_{325,\mathrm{lit}} = 47.9/49 = 0.98. \tag{12}$$

Magnetic susceptibility. The filled tube weighed 0.9090 g (Figure 11b). The datasheet gives two tare values. The tare of 0.8223 g gives the entered mass of 0.0867 g, and the tare of 0.8264 g gives 0.0826 g. The values below compare the displayed readings with the literature and with one unpaired electron. No experimental susceptibility or moment is derived.

$$\text{displayed} \quad \chi_v = -0.005 \times 10^{-4}, \qquad \chi_g = 0.000 \times 10^{-4}\ \mathrm{cm^3\,g^{-1}}, \tag{13}$$

$$\chi_g(\text{from } \chi_v) = \chi_v/\rho \le \frac{-5 \times 10^{-7}}{1.5} = -3.3 \times 10^{-7}\ \mathrm{cm^3\,g^{-1}} \quad (\text{display} \le -0.003 \times 10^{-4}), \tag{14}$$

$$\chi_{g,\mathrm{lit}} = \frac{-125 \times 10^{-6}}{186.04} = -6.7 \times 10^{-7}\ \mathrm{cm^3\,g^{-1}} \quad (\text{display } -0.007 \times 10^{-4}), \tag{15}$$

$$\chi_g(S = \tfrac{1}{2}) = \frac{\mu^2}{8TM} = \frac{1.73^2}{8 \times 296.15 \times 186.04} = +6.8 \times 10^{-6}\ \mathrm{cm^3\,g^{-1}} \quad (\text{display } +0.068 \times 10^{-4},\ T = 23\ ^\circ\mathrm{C}). \tag{16}$$

Appendix C. Calculated ESI isotope envelope for $C_{10}H_{10}Fe^+$

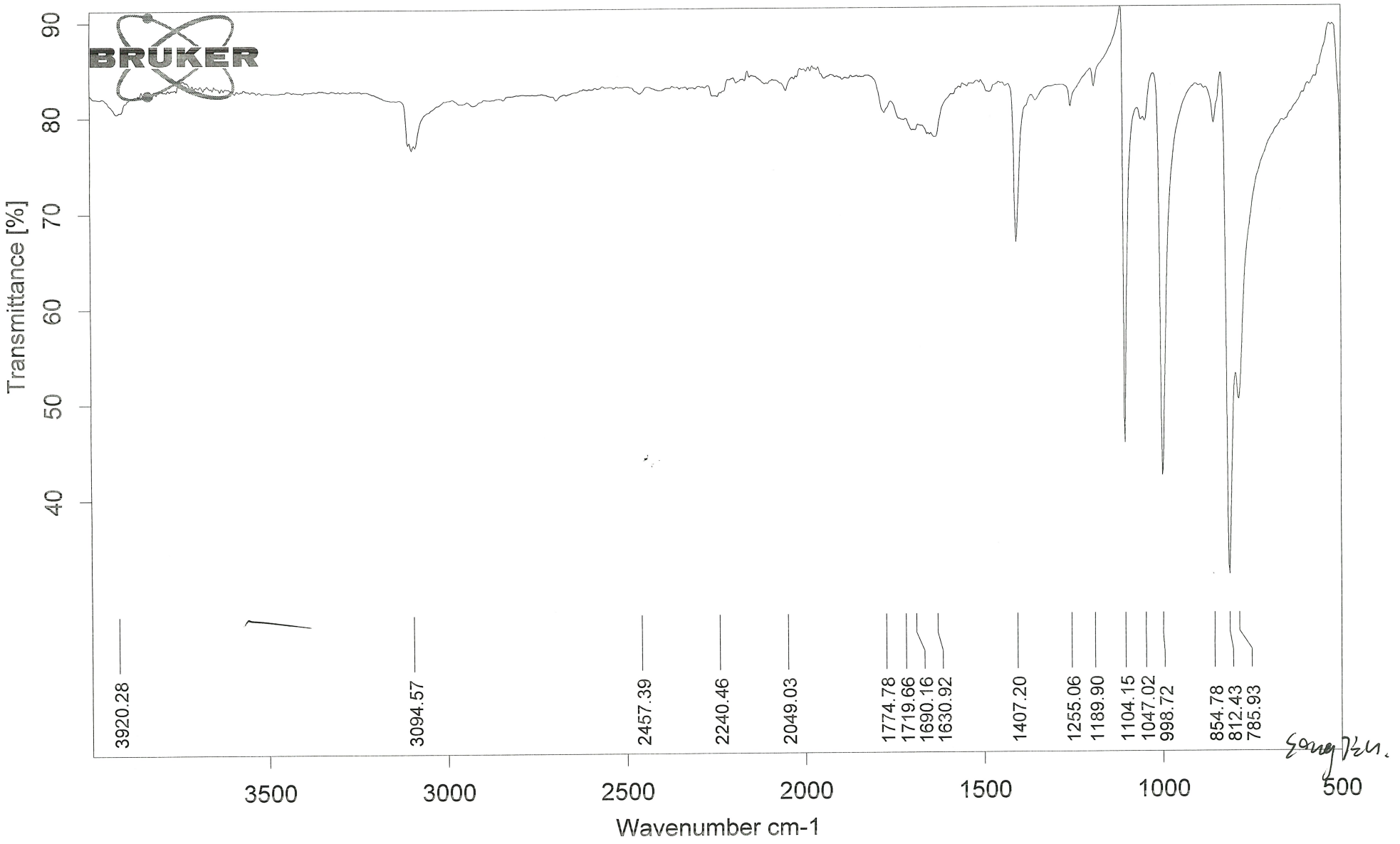
Natural-abundance distributions[9] were convolved for ten C, ten H and one Fe. The ^{56}Fe group at nominal $m/z = 186$ is normalised to 100 (singly charged ion, no adducts). Observed values were read from the instrument printout (base peak = 100).

Nominal m/z	Calculated / %	Observed / %	Origin
184	6.37	6.5	^{54}Fe
185	0.70	0.7	^{54}Fe + ^{13}C
186	100.0	100	$^{12}C_{10}\,^{1}H_{10}\,^{56}Fe$
187	13.24	14.2	^{13}C (10.7%) or ^{57}Fe (2.1%)
188	1.10	1.2	$^{13}C_2$, ^{58}Fe, or ^{13}C + ^{57}Fe
189	0.06	—	

As a first-order check, $I_{M+1}/I_M \approx 10\,x(^{13}\mathrm{C})/x(^{12}\mathrm{C}) + x(^{57}\mathrm{Fe})/x(^{56}\mathrm{Fe}) + 10\,x(^{2}\mathrm{H})/x(^{1}\mathrm{H}) \approx 0.108 + 0.023 + 0.001 = 0.132$.

Appendix D. Original IR, UV–Vis and ESI-MS printouts

The IR and UV–Vis pages are the signed originals, with the handwritten ε calculation on the UV–Vis page. The ESI-MS pages are the instrument export of 17 September 2026.

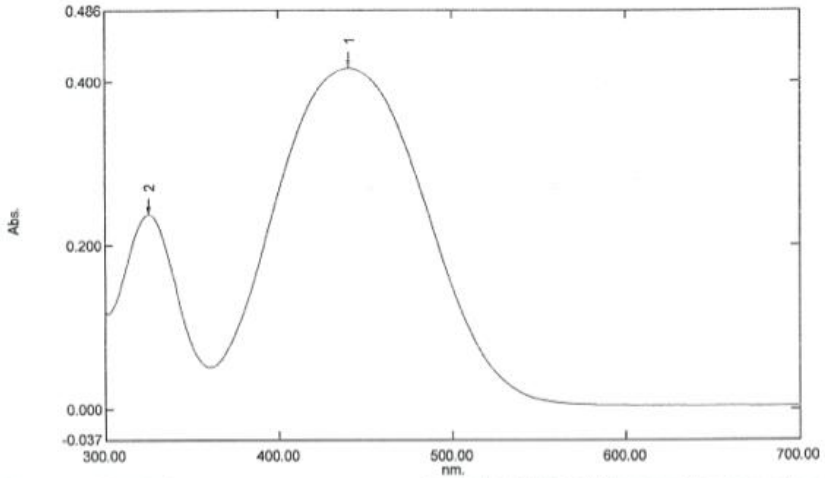


D:\CM3191\17Sep2026\Ferrocene_Kim_donghwan.0	Ferrocene_Kim_donghwan	Solid	17/9/2026

Spectrum Peak Pick Report

09/17/2026 03:09:58 PM

Data Set: KIM DONGHWAN_Ferrocene_150815.spc - RawData



No.	P/V	Wavelength	Abs.	Description
1	P	440.50	0.416	
2	V	325.00	0.239	

[Measurement Properties]
Wavelength Range (nm.): 300.00 to 700.00
Scan Speed: Fast
Sampling Interval: 0.5
Auto Sampling Interval: Enabled
Scan Mode: Auto

[Instrument Properties]
Instrument Type: UV-1800 Series
Measuring Mode: Absorbance
Slit Width: 1.0 nm
Light Source Change Wavelength: 340.0 nm
S/R Exchange: Normal

[Attachment Properties]
Attachment: None

[Operation]
Threshold: 0.0010000
Points: 4
InterPolate: Disabled
Average: Disabled

[Sample Preparation Properties]
Weight:
Volume:
Dilution:
Path Length:
Additional Information:

Soong Tae Kim

$M_{Ferrocene} = 186.04\ g/mol$

$c = m/(MV)$

$= \dfrac{0.0232\ g}{\left[186.04\ \frac{g}{mol}\right] \cdot 0.02500\ L}$

$= 4.988 \times 10^{-3}\ mol/L$

$A = \varepsilon c l \rightarrow A/cl = \varepsilon$

$\lambda_1 = 440.50\ nm$

$\varepsilon = \dfrac{0.416}{4.988 \times 10^{-3}\ mol\ L^{-1} \times 1.00\ cm}$

$= 83.4\ M^{-1} cm^{-1}$

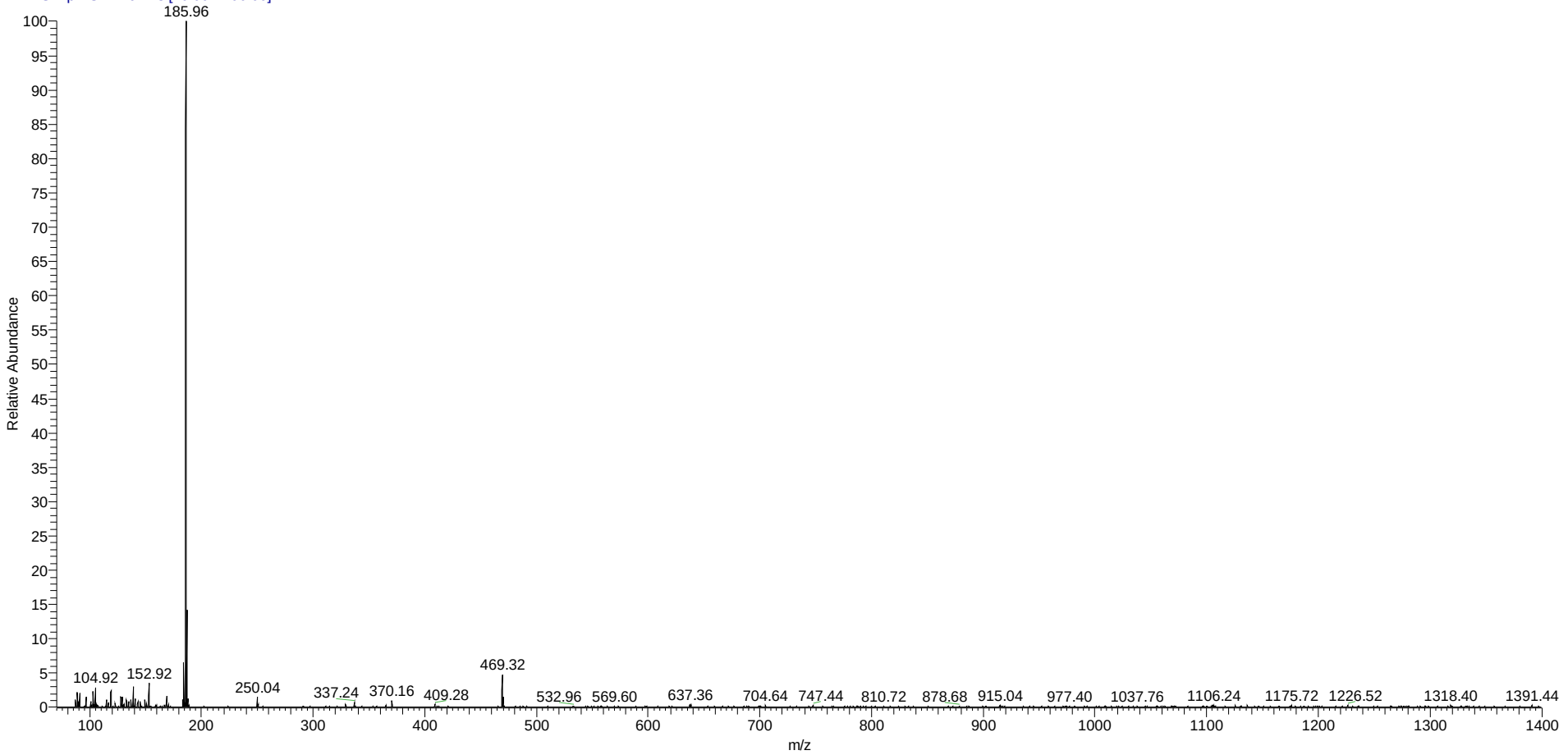
$\lambda_2 = 325.00\ nm$

$\varepsilon = \dfrac{0.239}{4.988 \times 10^{-3}\ mol\ L^{-1} \times 1.00\ cm}$

$= 47.9\ M^{-1} cm^{-1}$

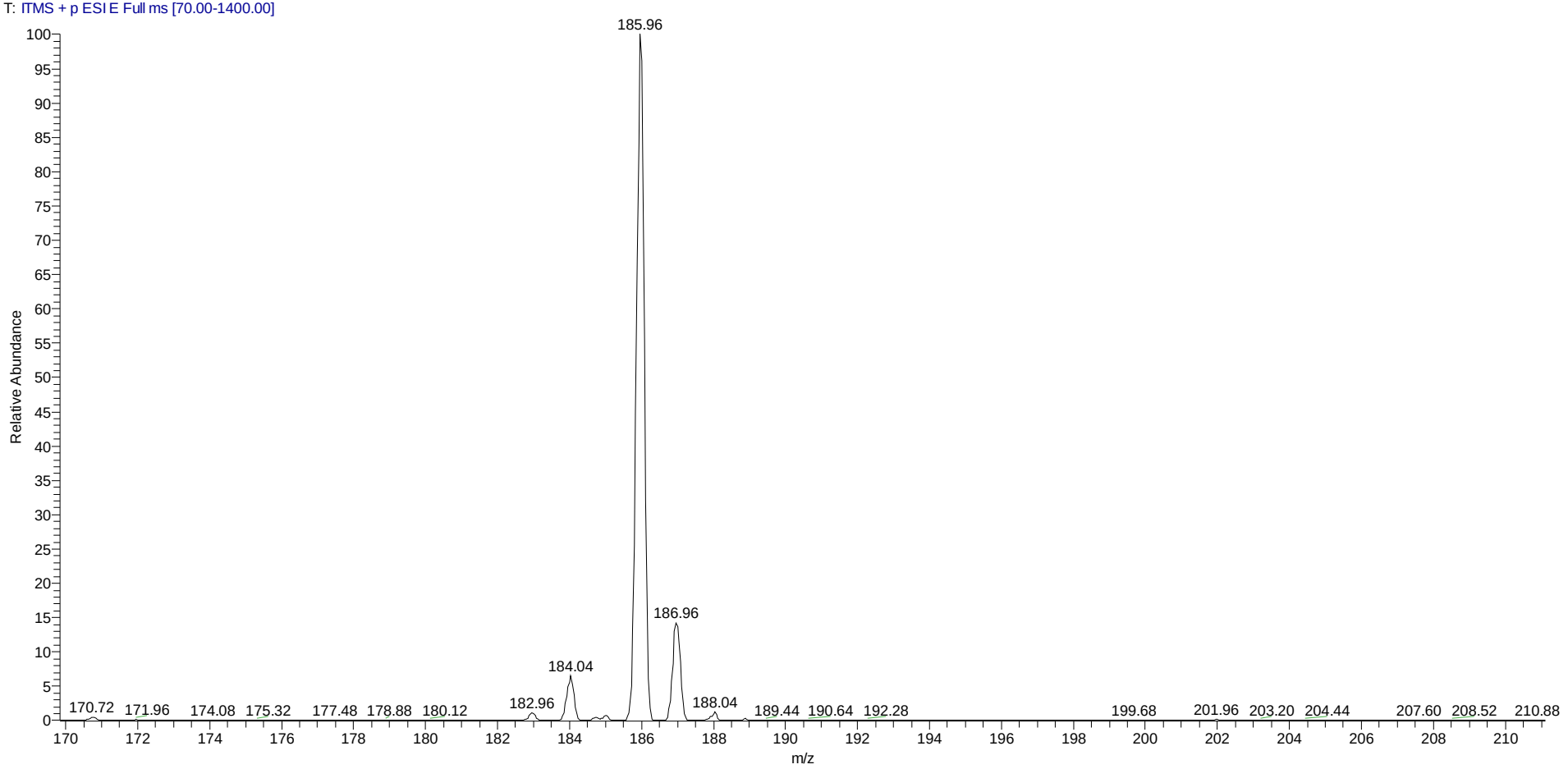
Z:\0_CM3191\20260917\SN9 17/9/2026 5:56:29 pm Enter Module Name
Enter Student Name

SN9 #30 RT: 0.51 AV: 1 NL: 1.41E5
T: ITMS + p ESI E Full ms [70.00-1400.00]



Z:\0_CM3191\20260917\SN9 17/9/2026 5:56:29 pm Enter Module Name
Enter Student Name

SN9 #30 RT: 0.51 AV: 1 NL: 1.41E5
T: ITMS + p ESI E Full ms [70.00-1400.00]



Appendix E. Signed datasheets

Week 5 datasheet (p. 4) and Week 6 datasheets (pp. 2, 4 and 5).

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 5

Session I: Synthesis and Microscopic Image of Crude Ferrocene

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

~30.038 ice

Reagent	Nominal	Actual g	Remarks
KOH pellets / anhydrous DME	4.0 g / 12 mL	4.0170 g	
Cyclopentadiene monomer	1.0 mL		
$FeCl_2 \cdot 4H_2O$ / DMSO (dissolve, rinse)	1.3 g / 7, 3.5 mL	1.3274 g	
Conc. HCl / ice	10 mL / 30 g		
DI water (rinse / wash 3×)	10 / 20 mL		

2. Conditions

	Nominal	Actual	Remarks
N_2 purge before CpH addition	3–5 min		
Stirring after CpH addition	10 min		
$FeCl_2$ addition rate / stirring after	20 drops min^{-1}, 30 min		
Stirring after acid quench	15–20 min		
Vacuum drying	~30 min		

3. Colour checkpoints — two of these tell you the reaction is working

Stage	Expected	Observed / time	✓
KOH in DME, before CpH	white undissolved solid, colourless liquid	(pale)	
10 min after CpH	**deep burgundy / red-brown.** A pale pink means deprotonation is incomplete	pinkish brown?	
$FeCl_2 \cdot 4H_2O$ in DMSO	**pale green.** Yellow or orange means Fe^{2+} is already oxidising		
During the $FeCl_2$ addition	darkens to a dark orange-brown slurry		
On quenching into ice-acid	orange precipitate; some gas evolution	green? + bright orange + some bubbling	
Filtrate and washings	pale yellow-green, fading to colourless		
Crude ferrocene, dried	orange to orange-brown powder		

The two bold rows are the checks worth making *before* committing the next reagent. A burgundy solution says the cyclopentadienide formed; a pale pink says stir longer. A green iron solution says the Fe^{2+} is still intact; if it has gone yellow or orange, oxidation has already cost you yield and the low result is explained. Write what you see, not what is printed.

4. Crude product

Mass of empty vial	1.5428.	N_2 flow maintained throughout ☐✓	ok
Mass of vial + crude	1.8127	Gas evolution on quenching ☐	slightly?
Mass of crude ferrocene	0.27??	Microscopic image taken ☑	a/r/orange.

To be completed at the end of the session		
Student signature	GTA name	GTA signature
		Song Yan 10/9/2026

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

2026/09/10

WEEK 6 (Mon. / Thu.)

INORGANIC EXPERIMENT 2 — SESSION II

Purification and Characterization of Ferrocene

Venue	General & Synthetic Chemistry Lab, Block S5 Level-1	**Instrument / technique**	Stereomicroscope, MSB, IR, UV–Vis, ESI-MS, ^{1}H NMR
Time	10:00 a.m. – 4:00 p.m. (Mon. or Thu.)	**Source**	Manual v0724 + lecture notes

(a) Purification by sublimation

1. **Weigh both larger petri dish covers empty** on the analytical balance and record each.
2. Spread the crude ferrocene on the smaller petri dish base; cover with a weighed cover.
3. Raise the hotplate *slowly*; **100 °C for cycle 1**, and never past **150 °C**. Place the 400 mL beaker of ice water on the cover.
4. Take the cover off the heat, let it cool, recover the sublimed ferrocene. Repeat until nothing further sublimes.
5. Weigh each loaded cover; transfer the product to a clean transparent bag or vial.

WHAT TO EXPECT FROM EACH CYCLE — SEE APPENDIX G

Cycle 1 mostly moves **water**: the crude is still damp from last week's filtration, so expect a wet, pale cover and little orange. Raise the temperature from cycle 2 or 3 and the orange flakes appear. Non-volatile impurities stay on the *base*, which darkens as the cycles proceed — that residue is the evidence that purification happened.
Let the cover cool before handling, invert it over the bag and not the bench, and re-weigh the cover empty before reusing it. Only two are issued.

(b) Characterization — six measurements

1. **Stereomicroscopy.** 14×, one image, matriculation number typed on it. Compare against the Week 5 crude image; this pair is the qualitative evidence for purification.
2. **Magnetic susceptibility.** Grind **0.09–0.12 g** in the agate mortar. Both χ_v and χ_g, five photographs. *Read Appendix H before computing a moment; Appendix I shows why the answer must be zero.*
3. **UV–Vis.** ~~35–40 mg~~ 20–25 mg weighed to four decimals, in methanol, **25 mL volumetric flask**. Record 300–700 nm; dilute until $A = 0.1$–1.0. Determine ε for each peak; the calculation must be handwritten on the spectra.
4. **IR.** Solid spectrum 500–4000 cm^{-1}, ATR or KBr. Record which. *Appendix C.*
5. **^{1}H NMR.** **~1 mg** into a tube containing **0.6 mL $CDCl_3$**. One sharp singlet near δ 4.15 for all ten protons is the result — the ten C–H are equivalent because both rings rotate freely and all five carbons of each bind iron equally.
6. **ESI-MS.** ~~1–2 mg~~ 5mg into a sample tube. *Appendix B.*

THE MSB PROCEDURE — FIVE PHOTOGRAPHS, ALL ASSESSED

Switch on and wait for a zero reading (−0.000E−5Z) [**photo 1**]. Weigh an empty tube, place it on the detector, read, press TARE, confirm zero. Remove it, introduce the ground sample, measure the packed length (**15.0–25.0 mm**) and mass (**0.0010–0.9999 g**). Press LENGTH and enter three digits [**photo 2, before the last digit**]. Press WEIGHT and enter [**photo 3, before the last digit**]. Press MAG SUS and record χ_v [**photo 4**] and χ_g [**photo 5**]. The display advances the instant the last digit is entered.

Before you leave: ☐ Purified ferrocene labelled ☐ Microscope image ☐ Five MSB photos ☐ IR and UV–Vis spectra ☐ ESI-MS and NMR submitted ☐ Datasheet signed

original sample: microscopy ✓ → one more for extra sample → 1.4447 g initially (with vial)
^{1}H NMR sample submission ✓
ESI-MS sample submission
ATR-IR
UV-Vis.

1.1083
− 1.0989
(extra 20 mg → new (extra sample

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 6 (1 of 2)

Session II: Sublimation Record

Full name (as on student ID)	Matriculation no.	Date
KEN DONGHUAN.	A0266 094K	2026/09/17.
Lab session ☐ Mon ☑ Thu	Fumehood no. 137.	S/N no. assigned by lab officer

1. Starting material

petri s : 58.9177 g

Mass of crude ferrocene from Week 5		Hotplate setting, cycle 1	
Cover A, empty (g) Left	48.6311	Cover B, empty (g) right.	48.7280

2. Sublimation cycles

extra 45.7688

extra 48.6345

Cycle	Set / °C	Cover	Cover + Fc (g)	Fc (g)	Appearance of cover / base
1	~100°C	A / B	48.6714	0.0403	} 0.0526 g total
2	~140°.	A / B	45.7843	0.0123	
3		A / B	45.7792	0.1104	
4		A / B	48.6948.	0.0563	
			Total purified ferrocene (g)		58.9315 − 58.9177 = 0.0138

new (extra)

extra: 0.1805 g

3. Colour checkpoints

Stage	Expected	Observed / time	✓
Crude, before heating	orange to orange-brown powder		
Cover after cycle 1	**wet and pale** — mostly condensed water, little orange		
Cover after cycles 2–3	orange flakes or plates		
Petri dish **base**, later cycles	darkens; non-volatile residue left behind		
Purified product	bright orange crystalline flakes		
After grinding for the MSB	light orange free-flowing powder		
Methanol solution for UV	clear yellow-orange		

Expected is a checkpoint, not an answer. A pale cover after cycle 1 is the normal result, not a failure; record what you see and the time.

To be completed at the end of the session		
Student signature	GTA name	GTA signature Sang Jun 17/09/26

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

2026/09/17 page 1.

1-vial: 0.0232 g

Submit extra sample vial too

extra : 0.3021 g.

after sublimation/purification :

empty vial : 1.0961 g

1.8845

0.0945

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 6 (2 of 2)

Session II: Characterization Record

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

4. Magnetic susceptibility — Sherwood MSB AUTO

Zero reading		Packed length L (mm)	17 mm
Empty tube mass (g)		Sample mass m (g)	0.0867
Volume susceptibility χ_v	-0.005	Mass susceptibility χ_g	0.222
Room temperature (K)		Sign of χ_g ☐ − ☑ 0 ☐ +	

Photographs: ☐ zero ☐ before last digit of length ☐ before last digit of weight ☐ χ_v ☐ χ_g **A reading of exactly zero is a resolution limit, not a moment — Appendix H.**

5. UV–Vis dilute until $A = 0.1 \sim 1.0$. → look for ε.

Mass weighed (g, four decimals)		Path length (cm)	1.00
Flask volume (mL) / solvent	25, MeOH	Concentration c (M)	
Dilution factor, if any		λ_1 (nm) / A_1	
Expected λ: ~440 and ~325 nm		λ_2 (nm) / A_2	

6. IR, NMR, ESI-MS and imaging

IR method ☑ ATR ☐ KBr		Mass for ^{1}H NMR (~1 mg)	
Microscope image of purified ☐		Mass for ESI-MS (1–2 mg)	
S/N no. for NMR		S/N no. for ESI-MS	

7. Yield and observations

Purified mass / crude mass, and the % recovered	Crude vs purified under the microscope — size, habit, transparency

To be completed at the end of the session		
Student signature	GTA name	GTA signature
		Cheng Zhen 17/09/26

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

2026/09/17 page 12

photo - control unit and magnetic susceptibility scale #.

17 mm.

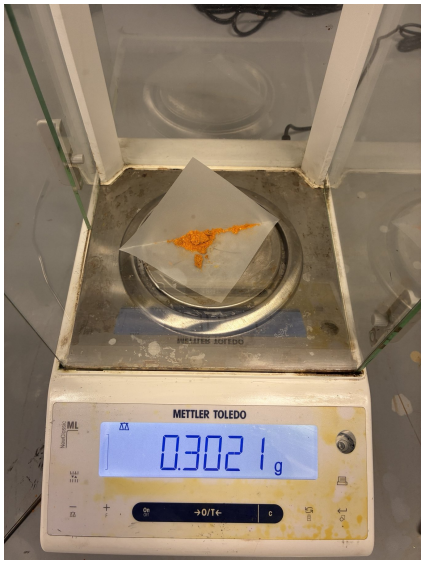
1. zero
2. ~~control~~ unit (7) length (2)
3. weight. (8)
4. control (9)
5. balance. (9)

total 5 photos.

0.8988
0.8223
0.0765 g.

0.9040
- 0.8264
0.0776.

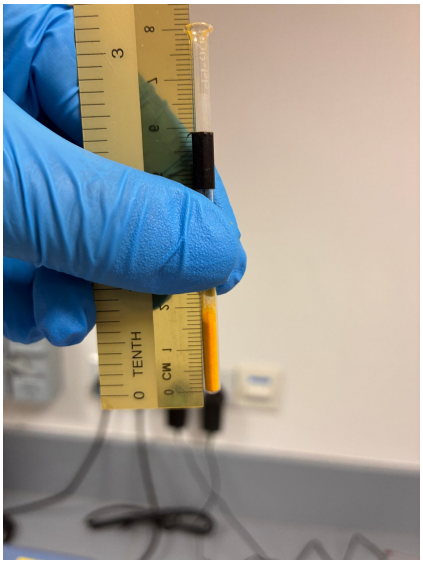
Appendix F. Weighing and magnetic-balance photographs



(a) second portion of crude ferrocene on tared weighing paper, 0.3021 g



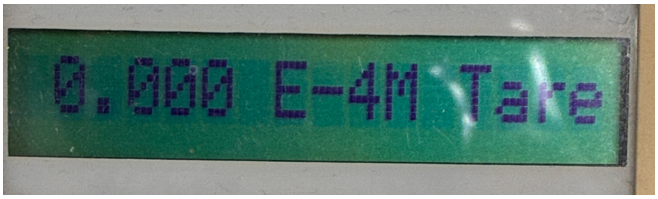
(b) filled MSB tube on the balance, 0.9090 g



(c) packed sample beside the ruler, 17 mm

Figure 11. Weighing and packed-length photographs.







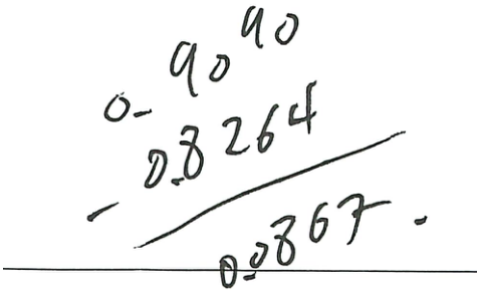


Figure 12. Enlarged display crops of the volume-mode result, the mass-mode result and the zero-reading frame, with the final handwritten mass subtraction on the datasheet (Appendix B).